

EL YOUSFI HAJAR

NETWORKS & TELECOMS ENGINEERING STUDENT (M2)

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Availability : 6-month Final-Year Internship (PFE) starting February 2027 — Casablanca / Rabat

PROFESSIONAL SUMMARY

Master's student in ISERT engineering (Networks & Telecoms, FSTM) with strong expertise across the telecommunications ecosystem from 2G to 5G (CDMA, O-RAN, xApp) and AI applied to networks. Rigorous and passionate about innovation, seeking a 6-month Final-Year Internship (PFE) in networks / telecommunications.

EDUCATION

FSTM – Master's in Embedded Systems, Networks and Telecommunications Engineering (ISERT)	2025 – 2027
• 2nd year (M2): networks & telecoms, embedded systems, artificial intelligence.	
FSTM – Bachelor's in Computer Science and Multimedia Networks (IRM)	2024 – 2025
FSTM – DEUST in Computer Science	2022 – 2024
Ain Harrouda High School – Baccalaureate in Physical Sciences (French Option)	2021 – 2022

INTERNSHIP

Proactive Load Balancing preserving QoE using Deep Learning (O-RAN 5G) FSTM	Jun – Aug 2026
• Predictive load-balancing xApp for the O-RAN Near-RT RIC: LSTM/GRU prediction (92% accuracy) + DRL agent (PPO), with handover decisions through the E2 interface (E2SM).	
Design of a Smart Intrusion Detection Camera (Raspberry Pi, Embedded Deep Learning) FSTM	Jun – Aug 2026
• Real-time people detection on Raspberry Pi 4 (PFA): YOLOv8n, 3-level alerts, GPIO (LED/buzzer) and MQTT publishing.	
MENPS – IT Directorate (Bachelor Internship) Rabat	Apr – Jun 2025
• Development of the Mostaqbali application (React.js, Spring Boot) for the Ministry of National Education: application management, tracking, validation, securing data flows and automating the administrative process.	

PROJECTS

Edge AI Telecom Gateway – Predictive maintenance of telecom sites Python, FastAPI, WebSocket	2026
• Predictive maintenance of cellular towers: real-time analysis of telemetry (temperature, voltage, vibration) at the edge.	
• Anomaly detection (Isolation Forest) + FastAPI REST API and WebSocket streaming; deployable on edge machines.	
Cellular Network Planning and Optimization Atoll	2026
• Radio planning of a cellular network with Atoll: propagation modeling, dimensioning and coverage analysis.	
• Capacity and interference optimization; multi-band 4G/5G scenarios.	
IP Network Automation and Monitoring Platform Python, Grafana	2026
• Automation and monitoring platform for an IP network: metrics collection and automated configuration of network equipment.	
• Grafana dashboards and real-time alerting for availability and performance monitoring.	

TECHNICAL SKILLS

• Telecoms & RF : 2G, 3G, 4G, 5G, LTE, CDMA (BER/SNR), FTTH, GPON, Li-Fi, RF mixer 2.4 GHz CMOS	• AI & Data : Python, PyTorch, scikit-learn, OpenCV, LSTM/GRU, PPO, Anomaly Detection, Predictive Maintenance
• Mobile Networks & O-RAN : Near-RT RIC, xApp, E2, O-RAN, handover, QoE, Traffic Prediction	• Tools & Simulation : MATLAB, Simulink, ADS, CST, Atoll, Packet Tracer, Docker, Grafana, Git, LaTeX
• Data Networks : TCP/IP, VLAN/Trunk, STP, QoS, ACL, OSPF, Linux, GNS3	• Security : Wireshark, Firewall, IDS, VPN

CERTIFICATIONS

• CCNA Network Security – Cisco	• Claude AI – Anthropic
• CCNA Introduction to Network Security – Cisco	• Microsoft AI Classroom – Microsoft
• Network Defense – Cisco	

LANGUAGES & ACTIVITIES

- **French** : Fluent, **English** : Fluent, **Arabic** : Native, **Italian** : Beginner, **Spanish** : Beginner, **Chinese** : Beginner
- **Activities** : Open Data Club (open data and analysis projects, FSTM 2026); Robothings Club (robotics and embedded electronics, workshops and tech competitions, FSTM 2023 – Present).